

Meeting Log — Smart Scientific Calculator

Repository: <https://github.com/Krishnarajss/Smart-Scientific-Calculator>

The project was developed within a 1-week schedule (Day 1 to Day 7). The team held three meetings: a kick-off meeting on Day 1, a mid-project progress review on Day 4, and a final integration and submission review on Day 6.

Team Members

Name	Reg. No.	Role / Responsibility
Siddhartha Tiwari	20243275	Team Leader — Full-stack implementation (all iterations)
Krishnaraj Singh	20243278	SRS, Architecture, UML Diagrams, Use Cases, Repository Setup
Sourabh Verma	20243279	Testing (Test Cases & Evidence), Project Management Docs
Umar Rashid	20243591	Demo Video, Mid-Semester Project Report

Meeting Overview

Meeting	Planned Day	Purpose
Meeting 1	Day 1	Kick-off: topic, stack, requirements, roles and iteration plan
Meeting 2	Day 4	Progress review: demo of Iterations 1-5, Iteration 6 and testing plan
Meeting 3	Day 6	Final integration demo, test results review and submission packaging

Meeting 1 — Kick-off, Requirements & Planning

Field	Detail
Date	Day 1 — [Fill in date]
Time	[Fill in: e.g. 2:00 PM – 3:00 PM]
Mode	[Online / In-person]
Attendees	Siddhartha Tiwari, Krishnaraj Singh, Sourabh Verma, Umar Rashid
Chair	Siddhartha Tiwari (Team Leader)

Agenda

1. Project-topic selection and choice of technology stack.
2. Define the problem statement, user needs (UN-01 to UN-06) and functional and non-functional requirements.
3. Assign roles and plan Iterations 1-6 over the one-week schedule.

Discussion Summary

- The team decided to build a Smart Scientific Calculator using HTML/CSS/JavaScript for the frontend and Node.js + Express + MongoDB (via Mongoose) for the backend, following the incremental development model.
- Identified six user needs (UN-01 to UN-06), one set per iteration: basic arithmetic, scientific functions, error handling, responsive UI and database history.
- Functional requirements finalised for all six iterations. Non-functional requirements: performance (< 1 s response), usability and browser compatibility.
- Reviewed the proposed 3-tier architecture (Frontend → Express Server → MongoDB).
- Roles confirmed: Siddhartha Tiwari leads and implements all iterations; Krishnaraj Singh owns the repository and the requirements and design artifacts; Sourabh Verma handles testing and project-management documents; Umar Rashid handles the demo video and mid-semester report.

Decisions Made

- Project topic, incremental model and technology stack finalised.
- Iteration 1 scope: basic arithmetic operations (+, -, *, /, %, AC, DEL).
- Scientific functions ($\sqrt{}$, sin, cos, x^y) will use degree-based input, not radians.
- The GitHub repository is owned and set up by Krishnaraj Singh; documentation is kept on a separate SRS branch and code is merged through pull requests.

Action Items

Action	Owner	Deadline
Create GitHub repository and add initial project description to README.md	Krishnaraj Singh	Day 1
Write SRS document and finalise architecture diagram	Krishnaraj Singh	Day 1
Prepare use-case descriptions and user stories	Krishnaraj Singh	Day 2
Implement Iteration 1 (basic arithmetic)	Siddhartha Tiwari	Day 1
Implement Iterations 2 & 3 (scientific functions)	Siddhartha Tiwari	Day 2
Implement Iteration 4 (error handling)	Siddhartha Tiwari	Day 3

Meeting 2 — Progress Review & Iteration 6 / Testing Plan

Field	Detail
Date	Day 4 — [Fill in date]
Time	[Fill in: e.g. 2:00 PM – 3:00 PM]
Mode	[Online / In-person]
Attendees	Siddhartha Tiwari, Krishnaraj Singh, Sourabh Verma, Umar Rashid
Chair	Siddhartha Tiwari (Team Leader)

Agenda

1. Demo of Iterations 1–5 (arithmetic, scientific functions, error handling, UI).
2. Plan Iteration 6 (Express server, MongoDB and history API).
3. Plan test-case design, UML diagrams, project-management documents, report and demo video.

Discussion Summary

- Siddhartha Tiwari demonstrated Iterations 1–5: all basic operations, $\sqrt{}$ /sin/cos/ x^y , and the responsive grid-based UI with real-time display and button feedback worked as expected.
- Error handling covers division by zero, negative square root (invalid domain) and malformed expressions, shown to the user as a friendly “Error” message.
- Discussed the MongoDB schema { expression, result, createdAt } and the REST endpoints POST /api/history and GET /api/history.
- Sourabh Verma proposed grouping test cases by iteration so that every feature is covered; Krishnaraj Singh confirmed the UML class and sequence diagrams will follow the final code structure.

Decisions Made

- Iterations 1–5 accepted as demo-ready.
- Use Mongoose for schema validation and MongoDB interaction.
- History retrieval is limited to the 10 most recent records, sorted by createdAt in descending order.
- A total of 19 test cases will be written, covering all six iterations, with evidence and screenshots recorded in a separate document.

Action Items

Action	Owner	Deadline
Implement Iteration 5 UI polish and Iteration 6 (Express server, MongoDB, history API); connect frontend to API	Siddhartha Tiwari	Day 5
Prepare UML class and sequence diagrams	Krishnaraj Singh	Day 5
Design the 19 test cases and start the test-evidence document	Sourabh Verma	Day 5
Prepare project plan and meeting log	Sourabh Verma	Day 5
Draft the mid-semester report outline and demo-video script	Umar Rashid	Day 5

Meeting 3 — Final Integration Review & Submission Planning

Field	Detail
Date	Day 6 — [Fill in date]
Time	[Fill in: e.g. 2:00 PM – 3:00 PM]
Mode	[Online / In-person]
Attendees	Siddhartha Tiwari, Krishnaraj Singh, Sourabh Verma, Umar Rashid
Chair	Siddhartha Tiwari (Team Leader)

Agenda

1. Full integration demo of all six iterations, including database history.
2. Review test results and test evidence.
3. Review report, demo video, repository status and remaining work.
4. Plan final ZIP packaging and submission.

Discussion Summary

- Full end-to-end demo completed: basic arithmetic → scientific functions → error handling → UI → database history. All six iterations work together.
- Sourabh Verma reported that all 19 test cases passed: TC-001 to TC-006 (basic arithmetic), TC-007 to TC-010 (scientific functions), TC-011 to TC-013 (error handling), TC-014 to TC-015 (UI and interaction) and TC-016 to TC-019 (database history).
- Repository status checked: code, requirements and design artifacts are on GitHub, while the testing and project-management files (Sourabh Verma) and the report and demo files (Umar Rashid) are still local and need to be pushed.
- Umar Rashid presented the finished mid-semester report and the 5–7 minute demonstration video, which will be hosted externally with its link stored in DEMO_VIDEO_LINK.txt to stay within the upload limit.
- Known limitations and remaining work discussed: replacing eval() with a safe expression parser, keyboard input support, automated Jest tests, an in-page history panel, cloud deployment and usability testing.

Decisions Made

- All six iterations accepted; the project is complete for mid-semester submission.
- Contribution is recorded as equal (25% each) and each member will confirm the Contribution Record declaration.
- Remaining items (eval() replacement, keyboard input, Jest tests, history panel, deployment) are deferred to the post-mid-sem phase.

Action Items

Action	Owner	Deadline
Push testing and project-management files (05_Testing, 06_Project_Management) to the repository	Sourabh Verma	Day 6
Push report and demo files (01_Report, 07_Demo) and upload the video link	Umar Rashid	Day 6
Final check of Readme.md, package.json and application run instructions	Siddhartha Tiwari	Day 6
Verify repository structure and package the final ZIP for submission	Krishnaraj Singh	Day 7
Sign the contribution declaration; final review and submission	All members	Day 7

Reminder: Replace all [Fill in] placeholders (date, time and mode) with the actual details of each meeting before submission.